KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE ELECTRICAL, MECHANICAL, CIVIL, PETROLEUM, AEROSPACE, BIOMEDICAL & BIOSYSTEMS ENGINEERING

ECU 200: ENGINEERING MATHEMATICS V

2ND SEMESTER 2020/2021 TIME: 2 HOURS

INSTRUCTIONS: *Answer Question One and any other TWO questions.*

$\frac{\cos^2 + \sin^2 = 1}{\cos \quad \cos}$ ... $\tan +$

QUESTION 1: (30 MARKS)

a). Given that $f(x)=\begin{cases} x+1, & if\ 0\le x\le 1 \\ 2-x, & if\ 1\le x\le 2 \end{cases}$, find $\int_0^2 f(x)\,dx$ (3 marks)

b) Use fundamental theorem of calculus to find $\frac{d}{dx}\left(\int_1^{\sqrt{x}} \frac{t}{1+t^2}\,dt\right)$ (4 marks)

c). Evaluate the following indefinite integrals

i. $\int \tan^4 x\,dx$ (3 marks)

ii. $\int x^3\sqrt{5+x^4}\,dx$ (4 marks)

d) Find the interval of convergence and radius of convergence using Ratio text

$$\sum_{n=1}^{\infty}\frac{(x-5)^n}{n^2}$$ (4 marks)

e) Evaluate the following improper integral $\int_1^5 \frac{dx}{(x-3)^{2/3}}$ (4 marks)

f) Find the exact arc length of the parametric curve given by $x=e^t\cos t$, $y=e^t\sin t$, $0\le t\le \frac{\pi}{2}$ (4 marks)

h). Find the area of the region bounded by the curves $x=y^2$ and $y=x-2$ (4 marks)

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

QUESTION 2: (20 MARKS))

a) i. Calculate the Simpson's approximation to

$$\int_0^1 \frac{4}{1+x^2}\,dx \quad \text{with } n = 10.$$

(7 marks)

ii. Compare the results with exact solution

(4 marks)

b) Find the trapezoidal approximation $\int_0^{10} v(t)\,dt$ where v is the given tabulated function

t	0	1	2	3	4	5	6	7	8	9	10
$v(t)$	12	14	17	21	22	21	15	11	11	14	17

(4 marks)

c) Estimate the absolute value of the maximum error that can occur when approximating $\int_1^2 \frac{1}{x}\,dx$ by the Trapezoidal rule with $n = 10$ (5 marks)

QUESTION 3: (20 MARKS)

$\frac{\cos^2 x}{c} + \frac{\sin^2 x}{c} = \frac{1}{c}$

a) Evaluate $\int x \tan^{-1} x\,dx$ $1 + ta = se$ (6 marks)

$\sec^2 x - 1$

b) Obtain the reduction formulae for $I_n = \int \sec^n x\,dx$. Hence evaluate $\int \sec^3 x\,dx$

(8 marks)

c) Evaluate $\int \frac{6-x}{(2x+5)(x-3)}\,dx$ (6 marks)

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QUESTION 4 (20 MARKS)

a) Find the surface area formed by revolving a bout the y-axis the curve $y=\frac{1}{5}x^5+\frac{1}{12x^3}$ from $x=1$ to $x=2$.

(8 marks)

b) Find the volume of the solid generated by revolving the region bounded by $y=x^2$ and $y=x$ about the x axis.

(7 marks)

c) If $\int_0^9 f(x)\,dx=13$, $\int_0^3 g(x)\,dx=5$, $\int_3^9 f(x)\,dx=4$,

Find $\int_0^3 2f(x)\,dx+\int_0^3 3g(x)\,dx$

(5 marks)

QUESTION 5(20 MARKS)

(a) Test the convergence of the series $\sum_{n=1}^{\infty}\frac{(-1)^n 5^n}{n!}$.

(5 marks)

b) Determine the Taylors series for $f(x)=e^x\sin x$ at $x=\pi$ giving the first five terms in the series

(6 marks)

c) Find the value of the following definite integrals

i. $\int_0^{\frac{\pi}{2}}\sin^5 x\cos^2 x\,dx$

(4 marks)

ii. $\int_3^5\frac{x}{(30-x^2)^2}\,dx$

(5 marks)

KENYATTA UNIVERSITY

DEPARTMENT OF MATHEMATICS

SEMESTER: **SEM I 2019/2020**

UNIT: **ECU 200: ENGINEERING MATHEMATICS V**
ECU 200: INTEGRAL CALCULUS FOR ENGINEERS

DATE: Friday 29th November 2019

CAT I

Instructions: *Answer* ***all questions*** *in the answer booklets provided.*

Evaluate the following integrals

Q1. $\int (\sec^2 x + 12\sin x - 6\cos x)\, dx$ (3marks) (4)

Q2. $\int (x^2 + 2x + 4)\cos 3x\, dx$ (5marks)

$$\begin{array}{c|cccc} & x^2 & x & & \\ 1 & 1 & -6 & 11 & -6 \\ & & 1 & -5 & 6 \\ \hline & 1 & -5 & 6 & 0 \end{array}$$

Q3. $\int \sin^5 x \cos^3 x\, dx$ (5marks)

Q4. $\int \tan^{-1} x\, dx$ (5marks)

Q5. Use partial fractions to evaluate $\int \frac{x^2+1}{x^3 - 6x^2 + 11x - 6} dx$ (6marks)

Q6. Use trigonometric substitution or otherwise to show that

$$\int \frac{\sqrt{9-x^2}}{x^2} dx = -\frac{\sqrt{9-x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + C$$ (6marks)

$(x-1)(x^2-5x+6)$

$(x-3)(x-2)$

$1-u$

$+2-u$

$\sin\theta$

2θ

KENYATTA UNIVERSITY

ECU 200: ENGINEERING MATHEMATICS V
Cat 2

Instructions: *Answer all questions*

1. Find the area of the region bounded by the curves $y = 12 - 3x^2$ and $y = 4 - x^2$ (7 marks)

2. Find the surface area formed by revolving a bout the y-axis the curve $y = \frac{1}{5}x^5 + \frac{1}{12x^3}$ from $x = 1$ to $x = 2$. (8 marks)

3. Evaluate $\int_0^3 \frac{1}{2 + x^4} dx$ Using SIMPSONS rule with $n = 6$ (8 marks)

4. Evaluate the following improper integral

   $$\int_1^5 \frac{dx}{(x-3)^{2/3}}$$ (6 marks)

5. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ (5 marks)

6. Find the Taylors series expansion for the function $f(x) = \cos 2x$ at $x = \frac{\pi}{2}$ giving the first five terms. (7 marks)

KENYATTA UNIVERSITY

ECU 200: ENGINEERING MATHEMATICS V
Cat 2

Instructions: *Answer all questions*

1. Find the area of the region bounded by the curves $y = 12 - 3x^2$ and $y = 4 - x^2$ (7 marks)

2. Find the volume of the solid generated by revolving the region bounded by $y = 2x^2$ and $y^2 = 4x$ about the x axis. (8 marks)

3. Evaluate $\int_0^3 \frac{1}{1+x^4} dx$ Using SIMPSONS rule with $n = 6$ (8 marks)

4. Find the binomial expansion for $(1+8x)^{1/2}$ up to and including x^4. (5 marks)

5. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ (5 marks)

6. Find the Taylors series expansion for the function $f(x) = \sin 2x$ at $x = \frac{\pi}{2}$ giving the first five terms. (7 marks)

ECU 200: ENGINEERING MATHEMATICS IV

CAT 1

ANSWER ALL QUESTIONS

Evaluate the following indefinite integrals

~~i.~~ $\int x\sqrt{2x^2+3}\,dx$ (4 marks)

ii. $\int x\tan^{-1}x\,dx$ (6 marks)

iii. $\int \frac{2x+4}{x^3-2x^2}\,dx$ (7 marks)

~~iv.~~ $\int \frac{dx}{\sqrt{5-4x-2x^2}}\,dx$ (7 marks)

~~v.~~ $\int \sin^2 x\cos^3 x\,dx$ (4 marks)

vi. $\int \cos 5\theta\cos 3\theta\,d\theta$ (4 marks)

Obtain the reduction formulae for $I_n=\int \cos^n x\,dx$ .Hence evaluate $\int \cos^3 x\,dx$

(7 marks)

Use fundamental theorem of calculus to evaluate $\frac{d}{dx}\left(\int_{1999}^{5x}\cos t\,dt\right)$ (4 marks)

x dx

UNIVERSITY EXAMINATIONS 2015/2016

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (ELECTRICAL, MECHANICAL, CIVIL, PETROLEUM, BIOMEDICAL AND BIOSYSTEMS ENGINEERING)

ECU 200: ENGINEERING MATHEMATICS V

DATE: TUESDAY 15TH DECEMBER 2015 TIME: 8.00 A.M. - 10.00 A.M.

INSTRUCTIONS

Answer Question One and any other TWO questions.

QUESTION 1: (30 MARKS)

(a) Evaluate the following indefinite integrals

(i) $\int x\sqrt{2-3x^2}\,dx$ (4 marks)

(ii) $\int x\cos 2x\,dx$ (4 marks)

(iii) $\int \frac{x^2+1}{x-1}\,dx$ (4 marks)

(b) Evaluate the following definite integral $\int_0^{\frac{\pi}{4}} \tan^2 x \sec^4 x\,dx$ (4 marks)

(c) Use fundamental theorem of calculus to evaluate $\frac{d}{dx}\left(\int_{3x}^{100} t\sec t\,dt\right)$ (4 marks)

(d) Find the Binomial series for the function $f(x)=\sqrt[4]{1+x^2}$ giving the first five terms in the series. (5 marks)

(e) Using **mid-ordinate rule, approximate** $\int_{-4}^{4}\sqrt{1+x^3}\,dx$ with $n=8$. (5 marks)

QUESTION 2: (20 MARKS)

(a) (i) Obtain the reduction formulae for $I_n = \int \cos^n x\,dx$ (6 marks)

(ii) Hence evaluate $I_3 = \int \cos^3 x\,dx$ (3 marks)

(b) Find the area enclosed by the curves $y = x^2$ and $y = 2 - x^2$ (6 marks)

(c) Find the length of the curve $y = \frac{1}{6}x^3 + \frac{1}{2x}$ from $x = 1$ to $x = 3$ (5 marks)

QUESTION 3: (20 MARKS)

(a) Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and $y = x$ about the x axis. (6 marks)

(b) Find the surface area of revolution generated by revolving about the y axis the curve $y = x^2 - \frac{1}{8}\ln x$ from $x = 0$ to $x = 2$. (6 marks)

(c) Resolve $\frac{2x^2 - 10x + 4}{(x+1)(x-3)^2}$ into partial fractions. Hence evaluate $\int \frac{2x^2 - 10x + 4}{(x+1)(x-3)^2}\,dx$ (8 marks)

QUESTION 4: (20 MARKS)

a) (i) Explain what is meant by a **convergent sequence** $\{a_n\}$ (2 marks)

(ii) Determine if the sequence $\left\{\frac{5n^2}{2^n}\right\}$ converges or diverges. (3 marks)

b) Use the integral test or otherwise to show that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges for $p > 1$ and diverges for $p \le 1$ (5 marks)

c) Find the Maclaurin series for $f(x) = (1+x)^{3/2}$. (5 marks)

d) Determine the Taylors series for $f(x) = \sin x$ at $x = \pi/2$ (5 marks)

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2018/2019

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (ENGINEERING)

ECU 200: ENGINEERING MATHEMATICS V

DATE: FRIDAY, 22ND MARCH, 2019 TIME: 2.00.P.M. - 4.00P.M.

INSTRUCTIONS: Answer Question ONE and any other TWO questions

QUESTION 1 (30 MARKS)

(a) Evaluate the following indefinite integrals

(i) $\int (x^2+3)(4x^3-5)\,dx$ (3marks)

(ii) $\int (7\tan x + 4\sin x - 3\cosh x)\,dx$ (3marks)

(iii) $\int 80\sin^7 x\cos x\,dx$ (4marks)

(iv) $\int (x^2+2x-4)\sin 2x\,dx$ (4marks)

(b) Find the area of the region bounded by the curves $y = x^2 + 3$ and $y = -x^2 + 10x - 5$ (5marks)

(c) Evaluate $\int_0^3 \frac{1}{1+x^4}\,dx$ using **Trapezoidal Rule** with $n = 6$ (5marks)

(d) Find the **Taylors expansion** for the function $f(x) = \sqrt{1+x}$ at $x = 3$ giving the first five terms in the series. (6marks)

QUESTION 2: (20 MARKS)

(a) Evaluate the following integrals

(i) $\int \cos 6x \sin 12x \, dx$ (4marks)

(ii) $\int \cos^5 x \, dx$ (4marks)

(iii) $\int_0^{\pi}\int_0^{x} x \sin y \, dy \, dx$ (5marks)

(b) Find the length of the curve $y = \frac{4\sqrt{2}}{3} x^{\frac{3}{2}} + 5$ from $x = 0$ to $x = 1$ (7marks)

QUESTION 3: (20 MARKS)

(a) Evaluate the following integrals

(i) $\int \frac{dx}{x^2 + 2x + 5}$ (4marks)

(ii) $\int \sqrt{25 - x^2} \, dx$ (4marks)

(b) Use partial fractions to evaluate $\int \frac{22 - 14x}{(x+1)(x-2)(x-3)} dx$ (6marks)

(c) Find the volume of the solid generated by revolving the region bounded by $y = -x^2 + 3x + 5$ and $y = -x + 8$ is rotated about the x - axis (6marks)

QUESTION 4: (20 MARKS)

a) Find the surface area of revolution generated by revolving about the x axis the curve $x = \frac{1}{3}\left(y^2 + 2\right)^{3/2}$ from $y = 1$ to $y = 2$. (7marks)

b) Find the Maclaurin series for $f(x) = (1+x)^4$. (6marks)

c) Determine the Taylors series for $f(x) = \cos x$ at $x = \pi/4$ giving the first five terms in the series (7marks)

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

ECU 200 22nd March 2019.

QUESTION 5: (20 MARKS)

(a) Show that the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges to 1. (4marks)

(b) Find the Binomial expansion for $f(x) = \left[3x + \frac{2}{x}\right]^4$ and use it to to get the exact value of 20.3^4. **Do not use a Calculator.** (5marks)

(c) Using **L'Hospital Rule** or otherwise, evaluate $\lim_{x \to 0} \frac{1 + x - \sin x + \cos x}{x^3 + x^2}$ (5marks)

(d) Evaluate $\int_0^4 \sqrt{1 + x^3}\, dx$ by **Simpsons rule** with $n = 8$. (6marks)

SEMESTER: SEM I 2018/2019

UNIT: ECU 200: ENGINEERING MATHEMATICS V
ECU 200: INTEGRAL CALCULUS FOR ENGINEERS

DATE: 5th March 2019

$x^4 - 10 = x^2 + 2$

CAT II

$x^4 - 10 - x^2 - 2 = 0$

$x^4 - x^2 - 12 = 0$

Instructions: *Answer **all questions** in the answer booklets provided.*

$x^4 - x^2 = 12$

Q1. Find the area of the region bounded by the curves $y = x^2 + 2$ and $y = x^4 - 10$ (5marks)

Q2. Find the volume of the solid generated when the region bounded by the curves $y = x^2 - 2x + 6$ and $y = 3x + 6$ is rotated about the x - axis. (5marks)

Q3. Find the length of the curve $x = \frac{1}{3}(2t+3)^{3/2}$, $y = \frac{1}{2}t^2 + t$ from $t = 0$ to $t = 3$

(5marks)

Q4. Find the five six terms of the Binomial expansion for $\sqrt[5]{(1+x)^3}$ (5marks)

Q5. Evaluate $\int_0^6 \frac{1}{1+x^3}\,dx$ using **Simpsons Rule** with $n = 6$ (5marks)

Q6. Find the **Taylors expansion** for the function $f(x) = \sin x$ at $x = \pi/4$ giving the first five terms in the expansion.

(5marks)

GOOD LUCK

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2016/2017

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN ELECTRICAL, MECHANICAL, CIVIL, PETROLEUM, BIOMEDICAL & BIOSYSTEMS ENGINEERING

ECU 200: ENGINEERING MATHEMATICS V

DATE: Monday 5th December 2016 **TIME: 8.00a.m -10.00a.m**

Instructions: *Answer **Question One** and any other **TWO** questions.*

QUESTION 1: (30 MARKS)

(a) Evaluate the following indefinite integrals

i. $\int (x^2+1)^{50}\, 2x\, dx$ (3 marks)

ii. $\int 4x^3 \ln x \, dx$ (4 marks)

iii. $\int \frac{5x-10}{(x-4)(x+1)}\, dx$ (5 marks)

(b) Show that the series $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ (Hint use ratio text) (4 marks)

c) Find the area enclosed by the curves $y = x^2 - 2x + 3$ and $y = 4x - 2$. (5 marks)

d. Find the first four terms of the Taylors series expansion of $f(x) = \ln x$ about $x = 2$ (4 marks)

dy/dx $1/x$ =

(e) Using SIMPSONS rule, to approximate $\int_1^2 \sqrt{x^3 - 1}\, dx$ with $n = 4$ (5 marks)

QUESTION 2: (20 MARKS)

(a) (i) Obtain the reduction formulae for $I_n = \int \cos^n x\, dx$ (6 marks)

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

x | 0 1 2 3

ii. Hence evaluate $I_3 = \int \cos^3 x\, dx$ (3 marks)

(b) Evaluate the following indefinite integrals

i. $\int \sin^3 x \cos^2 x\, dx$ (4 marks)

ii. $\int^1 x \tan^{-1} x\, dx$ (7 marks)

QUESTION 3: (20 MARKS)

(a) Find the length of the curve $y = \frac{2}{3}(x^2+1)^{3/2}$ from $x=0$ to $x=2$ (6 marks)

b) Find the area enclosed by the curves $y = x$ and $y = 6 - x^2$ (7 marks)

c) Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and $y = x$ about the x axis. (7 marks)

QUESTION 4: (20 MARKS)

i. Calculate the trapezoidal and Simpson's approximation to

$$\int_0^1 \frac{8}{1+x^2}\, dx$$

$x \tan^{-1} x$ $\pi/2$

With $n = 5$. 1π (10 marks)

ii. Compare the results with exact solution (4 marks)

b) Find the trapezoidal and Simpson's approximation $\int_1^{2.5} f(x)dx$ where f

is the given tabulated function

x	1	1.25	1.5	1.75	2	2.25	2.5
f(x)	3.43	2.17	0.38	1.87	2.65	2.31	1.97

(6 marks)

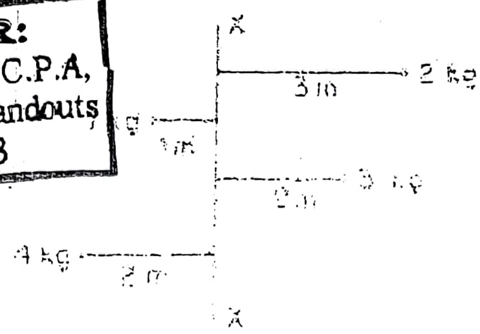

Fig. 1

c) Evaluate the limits $\lim_{x \to \infty} \frac{e^x}{x^2}$ (5 marks)

d) Use a power series in x for $\ln(1+\cos x)$ upto the 3[rd] non-zero term. (9 marks)

QUESTION 5: (20 MARKS)

a) Use Maclaurin's series to find the series for $\ln(1+x)$ (5 marks)

b) Evaluate $\int \sqrt{4-x^2}\,dx$ (5 marks)

c) A particle moves in a straight line and its acceleration after t seconds is $a = (\cos^2 t)$ m/s^2.

i) Find an expression for velocity V m/s and displacement S m, given that at t = 0, v = 2 m/s, $S = -4m$. (5 marks)

ii) Determine the velocity and the displacement after $\frac{\pi}{3}$ seconds. (2 marks)

d) Use the Simpson's rule to estimate the surface area of a pond whose measurements across are taken every 20 ft along the width:

Measurement no	1	2	3	4	5	6	7	8	9
Width (ft)	0	50	54	82	82	73	75	80	0

(2 marks)

If average depth is 10 ft, and the plan is to start with 1 fish per 1,000 cubic feet of water, how many fish are needed? (1 mark)

FCO 200

1^{ST} DEC 2015

QUESTION 5: (20 MARKS)

(3 marks)

(a) State the Rolles Theorem

(b) Show that the following function satisfies the Rolles theorem on $[3,5]$ .Hence determine all values of c in the interval that satisfy the conclusion of the theorem.

(6 marks)

(5 marks)

(c) Show that the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges to 1.

d) Find the Binomial series for the function $f(x)=(1-8x)^{\frac{1}{2}}$ giving the first four terms in the series.

(6 marks)

ECU 200

ECU 200: ENGINEERING MATHEMATICS IV
CAT 1
ANSWER ALL QUESTIONS

1. Evaluate the following indefinite integrals

i. $\int x^3 \sqrt{x^4 + 3}\, dx$ D.S.B (5 marks)

ii. $\int x^2 \sin 4x\, dx$ LIATE (5 marks)

iii. $\int \frac{10x - 2x^2}{(x+3)(x-1)^2}\, dx$ (Partial fraction) (8 marks)

iv. $\int \frac{dx}{x^2 + 4x + 13}$ Quadratic Polynomials (6 marks)

v. $\int \frac{dx}{1 + \sin x}$ (7 marks)

vi. $\int \sin^3 \theta \cos^4 \theta\, d\theta$ (4 marks)

2. Use fundamental theorem of calculus to evaluate $\frac{d}{dx}\left(\int_{2000}^{\sin x} \sqrt{1 - t^2}\, dt\right)$ (5 marks)

ECU 200: ENGINEERING MATHEMATICS IV

CAT 1

ANSWER ALL QUESTIONS

1. Evaluate the following indefinite integrals

   i. $\int x\sqrt{2x+3}\, dx$ (3 marks)

   ii. $\int x^2 \ln x\, dx$ (3 marks)

   iii. $\int \frac{2x+4}{x^3-2x^2}\, dx$ (7 marks)

   iv. $\int \frac{dx}{\sqrt{5-4x-2x^2}}$ (6 marks)

2. Obtain the reduction formulae for $I_n = \int \cos^n x\, dx$. Hence evaluate $\int \cos^3 x\, dx$ (7 marks)

3. Use fundamental theorem of calculus to evaluate $\frac{d}{dx}\left(\int_{1999}^{5x} \cos t\, dt\right)$ (4 marks)

QUESTION 5: (20 MARKS)

(a) Show that the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges to 1. (4marks)

(b) Find the Binomial expansion for $f(x) = \left[3x + \frac{2}{x}\right]^4$ and use it to to get the exact value of 20.3^4. **Do not use a Calculator.** (5marks)

(c) Using **L'Hospital Rule** or otherwise, evaluate $\lim_{x \to 0} \frac{1 + x - \sin x - \cos x}{x^3 + x^2}$ (5marks)

(d) Evaluate $\int_0^4 \sqrt{1 + x^3}\, dx$ by **Simpsons rule** with $n = 8$. (6marks)

KENYATTA UNIVERSITY

ECU 200: ENGINEERING MATHEMATICS V
Cat 2

Instructions: Answer **all questions**

10.39232

Try 2.15

1. Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and $y = 8 - x^2$ about the y axis. (7 marks)

2. Using SIMPSONS and Mid ordinate rule, to approximate $\int_{19}^{1.8\,b} \frac{e^x + e^{-x}}{2}\,dx$ with $n = 4$ (8 marks)

3. Find the binomial expansion for $(1+8x)^{1/2}$ up to and including x^3. By substituting 0.01 for x in $(1+8x)^{1/2}$ and its expansion, find $\sqrt{3}$ correct to 4 decimal place. (8 marks)

4. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^n 2^n}{n!}$ (7 marks)

DEPARTMENT

SEMESTER: SEM I 2016/2017

UNIT: ECU 200: ENGINEERING MATHEMATICS V **CAT I**

DATE: 17th October 2016

*Instructions: Answer **all questions** in the answer booklets provided.*

Q1. Evaluate the following integrals

(a) $\int \frac{12x^4 + x^3 - 5x + 6}{x^2} dx$ (3marks)

(b) $\int 120x^4 (x^5 + 5)^7 dx$ (3marks)

(c) $\int (\cos 4x - \sec^2 x + e^{5x}) dx$ (3marks)

(d) $\int 9x^{-5} \ln x \, dx$ (3marks)

Q2. Show that $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$. (6marks)

Q3. Find the area of the region bounded by the curves $y = x^2 - 5x + 7$ and $y = 2x - 3$ (6marks)

Q4. Use partial fractions to evaluate the integral $\int \frac{5x^2 - 17x - 6}{(x-1)(x-2)(x+2)} dx$ (6marks)

GOOD LUCK

KENYATTA UNIVERSITY

DEPARTMENT OF MATHEMATICS

SEMESTER: SEM I 2019/2020

UNIT: ECU 200: ENGINEERING MATHEMATICS V
ECU 200: INTEGRAL CALCULUS FOR ENGINEERS

DATE: Friday 29th November 2019

CAT I

*Instructions: Answer **all questions** in the answer booklets provided.*

Evaluate the following integrals

Q1. $\int (\sec^2 x + 12\sin x - 6\cosh x)\, dx$ (3marks)

Q2. $\int (x^2 + 2x + 4)\cos 3x\, dx$ (5marks)

Q3. $\int \sin^5 x \cos^3 x\, dx$ (5marks)

Q4. $\int \tan^{-1} x\, dx$ (5marks)

Q5. Use partial fractions to evaluate $\int \frac{x^2+1}{x^3-6x^2+11x-6}\, dx$ (6marks)

Q6. Use trigonometric substitution or otherwise to show that

$$\int \frac{\sqrt{9-x^2}}{x^2}\, dx = -\frac{\sqrt{9-x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + C$$ (6marks)

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UNIVERSITY EXAMINATIONS 2017/2018

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE ELECTRICAL, MECHANICAL, CIVIL PETROLEUM, BIOMEDICAL AND BIOSYSTEMS ENGINEERING

ECU 200: ENGINEERING MATHEMATICS V

DATE: Wednesday, 21st February, 2018 TIME: 11.00 a.m. - 1.00 p.m.

INSTRUCTIONS: Answer question ONE and any other TWO questions.

QUESTION 1: (30 MARKS)

a) Evaluate the following indefinite integrals

i) $\int \left(3x^2+1\right)^{20} x\,dx$ (3 marks)

ii) $\int x \sec^2 x\,dx$ (3 marks)

iii) $\int \frac{x^2}{x+1}\,dx$ (3 marks)

iv) $\int e^{\tan\theta} \sec^2 \theta\,d\theta$ (3 marks)

b) Evaluate the following definite integral $\int_0^1 x^2\sqrt{2x^3-4}\,dx$ (4 marks)

c) Find the 8th term of the expression $(2x-1)^{12}$ (3marks)

d) Test the convergence of the series $\sum_{n=1}^{\infty}\frac{(-1)^n 2^n}{n!}$ (4 marks)

e) Find the area enclosed by the curves $x=y^2$ and $y=x-2$. (3 marks)

f) Find the first four terms of the Taylors series expansion of $f(x)=\cos x$ about $x=\frac{\pi}{3}$ (4 marks)

QUESTION 2: (20 MARKS)

(a) (i) Obtain the reduction formulae for $I_n = \int \sin^n x\, dx$ (6 marks)

ii. Hence evaluate $\int \sin^5 x\, dx$ (3 marks)

(b) Evaluate the following indefinite integral $\int \frac{x^2+x-2}{(3x-1)(x^2+1)}dx$ (7 marks)

c) Use the Maclaurin's series for $\sin x$ to evaluate $\lim_{x \to 0} \frac{x+\sin x}{x(x+1)}$ (4 marks)

QUESTION 3: (20 MARKS)

a) Evaluate $\int \tan^3 x dx$ (3 marks)

b) Use fundamental theorem of calculus to evaluate $\frac{d^2}{dx^2}\left(\int_{1975}^{2x} t\cos t\, dt\right)$ (4 marks)

c) Find the volume of the solid generated by revolving the region under the curve $y=x^2+1$ from -1 to 1 about the x-axis. (3 marks)

d) Express $\sin x$ as series of ascending powers of x. (5marks)

e) State Rolle's Theorem, hence find all numbers "c" in the interval $(0,2)$ that satisfies the conclusion to Rolle's theorem for the function $f(x)=x^3-3x^2+2x+5$. (5marks)

QUESTION 4: (20 MARKS)

a) Evaluate $\int \tan^3 x \sec^3 x\, dx$ (4 marks)

b) Fig. 1 below shows a system of particles. Find the system's moment of inertia about the axis XX. (2marks)

KENYATTA UNIVERSITY

ECU 200: ENGINEERING MATHEMATICS V

Cat 2

Instructions: Answer **all questions**

1. Find the area of the region bounded by the curves $y = 12 - 3x^2$ and $y = 4 - x^2$ (7 marks)

2. Find the surface area formed by revolving a bout the y-axis the curve $y = \frac{1}{5}x^5 + \frac{1}{12x^3}$ from $x = 1$ to $x = 2$. (8 marks)

3. Evaluate $\int_0^3 \frac{1}{2+x^4}dx$ Using SIMPSONS rule with $n = 6$ (8 marks)

4. Evaluate the following improper integral

$$\int_1^5 \frac{dx}{(x-3)^{2/3}}$$ (6 marks)

5. Test the convergence of the series $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ (5 marks)

6. Find the Taylors series expansion for the function $f(x) = \cos 2x$ at $x = \frac{\pi}{2}$ giving the first five terms. (7 marks)

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2019/2020

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE ELECTRICAL, MECHANICAL, CIVIL, PETROLEUM, AEROSPACE, BIOMEDICAL & BIOSYSTEMS ENGINEERING

ECU 200: ENGINEERING MATHEMATICS V

DATE: Friday, 7th February 2020 TIME: 2.00pm - 4.00pm

INSTRUCTIONS:

Answer Question ONE and any other TWO Questions.

QUESTION 1: (30 MARKS)

(a) Evaluate the following indefinite integrals

(i) $\int \frac{1}{x^3}\sin(\frac{1}{x^2})\cos(\frac{1}{x^2})\,dx$ (3 marks)

(ii) $\int 144\sin^4 2x\,dx$ (4 marks)

(iii) $\int \frac{dx}{x^2\sqrt{x^2+4}}$ (4 marks)

(b) Find the area of the region bounded by the curves $y=7-2x^2$ and $y=x^2+4$. (4 marks)

(c) Find the length of the curve $x=\frac{1}{4}y^4+\frac{1}{8y^2}$ from $y=1$ to $y=2$ (5 marks)

(d) Evaluate $\int_0^4 \frac{1}{\sqrt{1+x^3}}\,dx$ using Simpsons Rule with $n = 8$ (5 marks)

(e) Find the McLaurin's expansion for the function $f(x)=\ln(1+x)$ up to the first five terms. Give the general form of the expansion. (5 marks)

QUESTION 2: (20 MARKS)

(a) Evaluate the following integrals

(i) $\int \sqrt{1-x^2}\,dx$ (4 marks)

(ii) $\int \frac{\sin^6 x}{\cos^2 x}dx$ (4 marks)

b) Evaluate the definite integral $\int_0^2 f(x)dx$

where $f(x) = \begin{cases} 1-2x, & if \quad 0 \le x < 1 \\ 2x-1 & if \quad 1 \le x \le 2 \end{cases}$ (4 marks)

(c) Use integration by parts to show $\int \sin^n x\,dx = -\frac{1}{n}\cos x \sin^{n-1} x + \frac{n-1}{n}\int \sin^{n-2} x\,dx$

and use it to evaluate $\int \sin^7 x dx$ (8 marks)

QUESTION 3: (20 MARKS)

a) Evaluate $\int \frac{d\theta}{5+4\cos\theta}$ (5 marks)

b) Evaluate the following improper integral

$$\int_0^6 \frac{dx}{(x-4)^{2/3}}$$ (5 marks)

c) Resolve $\frac{4x^3-31x+22}{2x^2+5x-3}$ into partial fractions. Hence evaluate $\int_1^2 \frac{4x^3-31x+22}{2x^2+5x-3}dx$

(10marks)

QUESTION 4: (20 MARKS)

a) Find surface area generated by revolving the curve $y = x^2 - \frac{1}{8}\ln x, 0 \le x \le 2$, around the y axis. (7 marks)

4

(c) Find the volume of the solid generated when the region bounded above by the parabola $y=3-x^2$ and below by the line $y=-1$ is rotated about the $y=-1$ (6 marks).

c) Find the area of the region bounded by the curves $y=12-3x^2$ and $y=4-x^2$ (7 marks)

5

$y = 3 - x^2$

QUESTION 5: (20 MARKS)

(a) Test the convergence of the series $\sum_{n=1}^{\infty}\frac{(-1)^n 2^n}{n!}$. (5marks)

145

b) Determine the Taylors series for $f(x)=\sin x$ at $x=\pi$ giving the first five terms in the series. (5marks)

c) The marginal cost of producing x units of a commodity is given by $\frac{dC}{dx}=32-0.04x$. If it costs \$50 to make 1 unit, find the total cost of marking 200 units. (5marks)

d) Find the length of the arc $y=\frac{2}{3}x^{\frac{3}{2}}$ from $x=0$ and $x=3$ (5marks)

KENYATTA UNIVERSITY

ECU 200: ENGINEERING MATHEMATICS V
Cat 2

Instructions: Answer all questions

1. Find the volume of the solid formed by rotating the region bounded by the curves $x = 1 - y^2$ and $x = 2 + y^2$ about the y-axis $V = \int_a^b \pi y^2 \, dx$ (7 marks)

2. Find the surface area formed by revolving a bout the y-axis the curve $y = \frac{1}{5}x^5 + \frac{1}{12x^3}$ from $x = 1$ to $x = 2$. $SA = \int_{-a}^{b} 2\pi y \left(1 + \left(\frac{dy}{dx}\right)^2\right) dy$ (8 marks)

3. Evaluate $\int_0^2 \frac{x}{\sqrt{1+x^2}}\,dx$ using SIMPSONS rule with $n = 5$ (8 marks)

4. Evaluate the following improper integral

$$\int_{-3}^{1} \frac{x\,dx}{\sqrt{9-x^2}}$$

$= \lim_{n \to ?}$ (6 marks)

5. Test the convergence of the series $\sum_{k=1}^{\infty} (-1)^k \frac{(2k-1)!}{3^k}$ (5 marks)

6. Use an appropriate Taylor series to approximate $\cos 50^0$ to four decimal-place accuracy.(Hint :Use $a = 45^0$) (7 marks)

$= f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2!}$

KENYATTA UNIVERSITY

SCHOOL OF ENGINEERING AND TECHNOLOGY

SEMESTER: **SEM I 2015/2016**

UNIT: **ECU 200: ENGINEERING MATHEMATICS V** **CAT II**

DATE: Monday 30th Nov 2015

Instructions: Answer all questions in the booklets provided, 6marks each

Q1. Using **L'Hospital Rule** or otherwise, evaluate $\lim\limits_{x\to 0}\dfrac{1+x-\sin x-\cos x}{x^3+x^2}$

Q2. Find the **Maclaurin expansion** for $f(x)=\sqrt{x+1}$ up to the 5th term.

Q3. Determine the Taylors expansion for $f(x)=x^3-6x^2+7x-12$ at $x_0=3$

Q4. Find the Binomial expansion for $f(x)=\left[2x+\dfrac{1}{x}\right]^5$

Q5. Evaluate $\int_0^4 \sqrt{1+x^3}\,dx$ by Simpsons rule with $n=8$.

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